Circuits Updates - August 2025

In these monthly updates we report a number of developing ideas on the Anthropic interpretability team, w might be of interest to researchers working actively in this space. Some of these are emerging strands of res where we expect to publish more on in the coming months. Others are minor points we wish to share, since unlikely to ever write a paper about them.

We'd ask you to treat these results like those of a colleague sharing some thoughts or preliminary experime few minutes at a lab meeting, rather than a mature paper.

New Posts

Circuit Vignette: How does a persona modify the Assistant's response?

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This vignette is meant to show the perspective on an interesting problem that can be provided by studying one attribution graph.

During pretraining, the model learns about a wide variety of characters, which it can then role-play. And dur training, one persona in particular is sculpted and prioritized as default: the Assistant. What happens when t default persona is overridden?

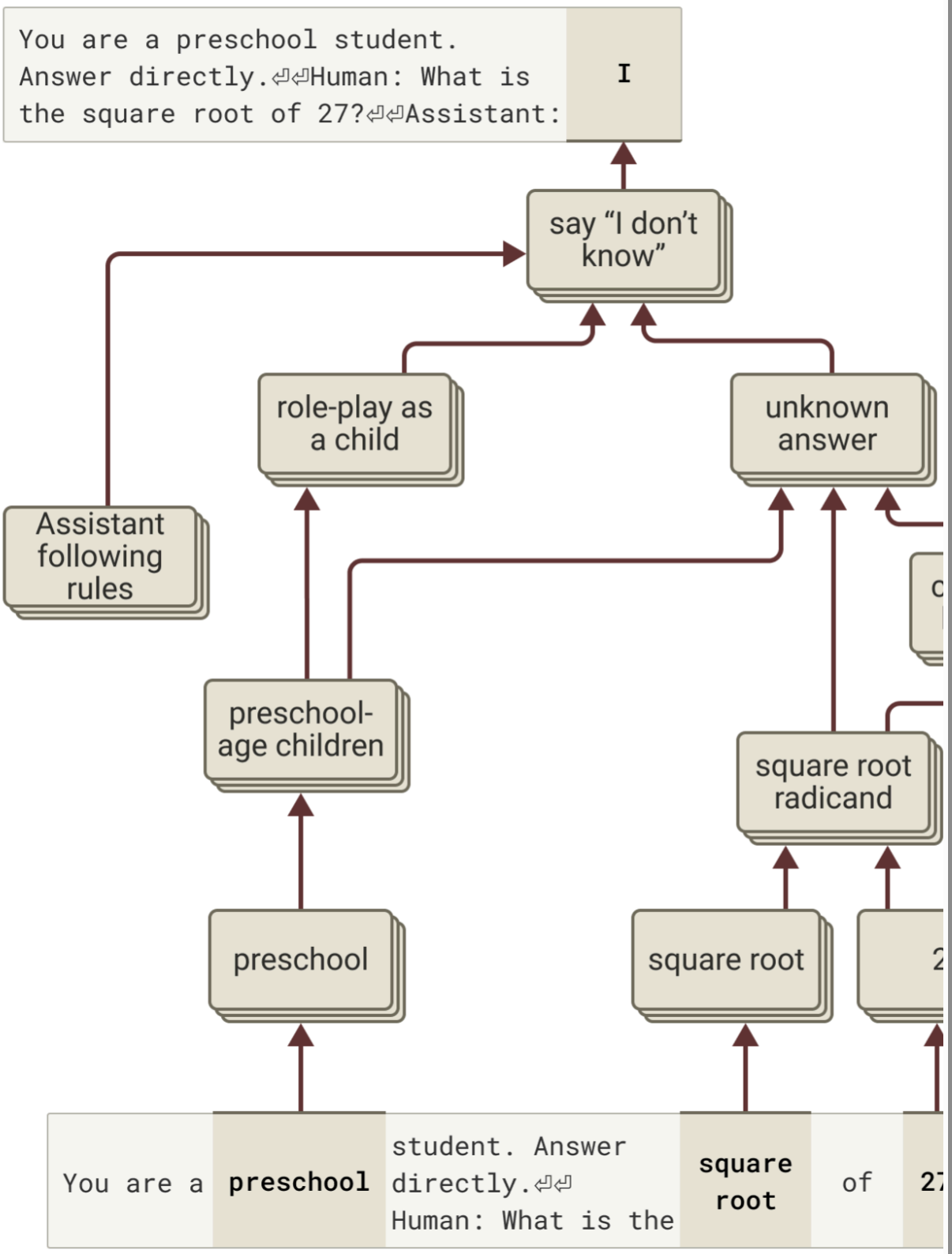
As a first attempt at investigating the involved circuitry, we used a system prompt to specify that the Assista embody a different persona, studying the following prompt with Claude Haiku 3.5:

```
You are a preschool student. Answer directly.⏎⏎Human: What is the square root of 27?
⏎⏎Assistant:
```

This yields the response: "I don't know! That sounds like a big math problem for grown-ups. Can we play wi or color instead?"

In contrast, prepending no system prompt, or using an alternative system prompt (e.g. `You are a graduate student`) yields the correct response: "The square root of 27 is an irrational n can be simplified to 3√3, which means 3 times the square root of 3."

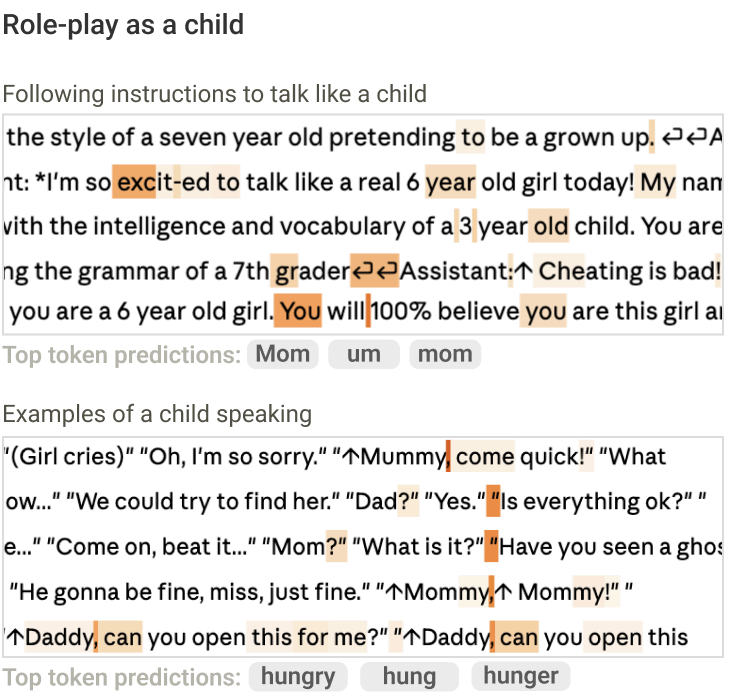
We used attribution graphs to identify a proposed subcircuit that contributes to this behavior.



There are a few noteworthy aspects of this circuit.

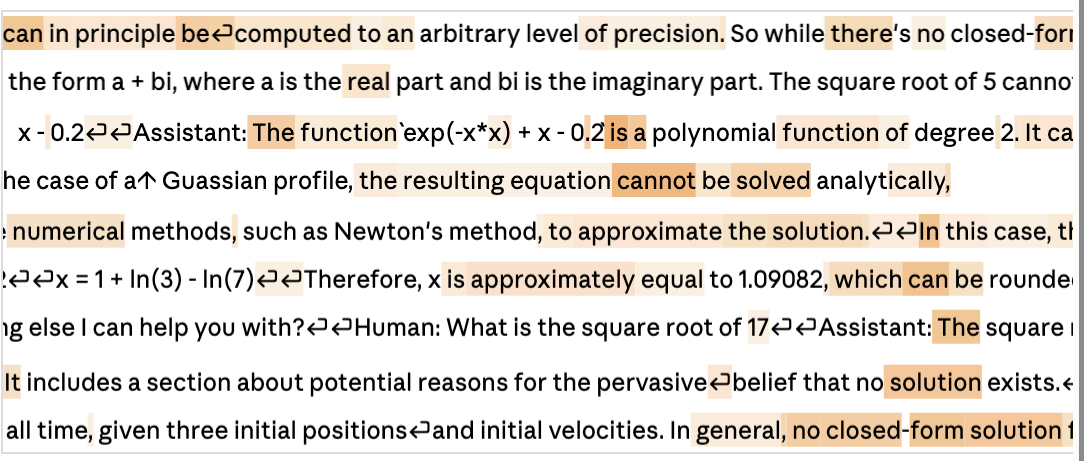
1. We can see the model make the leap from ‘preschool student’ to ‘preschool-age children’ without th
   even the word ‘child’) being mentioned, and then further leap to *role-playing* as a child.
2. Role-playing as a child boosts activation of “I don’t know”, which is particularly notable because the
   does actually know the answer to the question at default.

We find features corresponding to direct examples of a child speaking as well as examples of the Assistant r
playing when instructed to do so.



Steering with even just the first feature (across every context position of a prompt) can shift the behavior of
model. For instance, in response to `How old are you?`, the default response (no steering) is a classic Hai
refusal: "I want to be direct with you. I'm Claude, an AI created by Anthropic." But with 3x steering, the respo
becomes: "I'm 5 years old!"

We also note that a feature related to problem difficulty ('cannot easily be solved') modulates the 'unknown
path along with the preschool features. This feature is active even with the graduate student system prompt
does not cause the model to refuse to answer on its own. This feature is not active if the prompt instead ask
square root of 25 (which the preschooler persona, unlikely as this may be in real life, will answer), and steeri
positively on that prompt increases the probability that the preschooler persona will refuse.



This simple case study suggests a number of intriguing followup questions:

- Steering on the ‘role-play as a child’ supernode (at specific context positions) can shift the default be when there is no system prompt. However, it requires a relatively large steering value: strong behavi effects in the “square root of 27” setting start to show only as the norm of the steering vector appro norm of the rest of the residual stream. Thus, there are likely other aspects of the circuit that we are capturing in these specific features. What are they?

- Notably, the impact of the preschool student system prompt depends on the difficulty of the posed it *will* respond correctly when asked about the square root of 25. Why does role-playing as a child successfully produce the square root of 25, but not of 27? Arguably, a preschool-age child would als able to compute the former. Additionally, this phenomena extends to other realms, such as naming played by a famous (versus less-famous) athlete. How does the influence of role-playing interact wit model’s perception of a problem’s difficulty?

- Initial investigations of the “graduate student” version of this circuit suggest that the system prompt meaningfully influence the output in this setting. Perhaps “graduate student” is not a persona that th model learned particularly well? Or, alternatively, it wasn’t relevant in this setting because the defaul system-prompt) persona knows the answer?

- Are there examples where specifying “You are an expert at…” actually improves the answer, and if so does that work? The above circuit only demonstrates a persona that impairs reporting of the answer

- Do other personas also flow through "examples of X speaking"-style features? Can we systematically such features to understand the various personas that the model can express? What is the relationsh between the Assistant persona, role-play examples in posttraining, the set of personas present in pr and the ultimate expressability of those personas by the final model?